

Code:

```
#include <stdio.h> #include <stdlib.h>

#define OFFSET 200005

int hash_table[400010];

int main() { int M, Q, N; if (scanf("%d %d %d", &M, &Q, &N) != 3) return 0;

int A[N];
int j;
for (j = 0; j < N; j++) {
    scanf("%d", &A[j]);
}

// Difference array sweep across achievable ranges [A[i] - Q*M, A[i]
+ Q*M]
for (int i = 0; i < N; i++) {
    int left = A[i] - Q * M;
    int right = A[i] + Q * M;
    if (left < 0) left = 0;
    if (right > 400000) right = 400000;
    hash_table[left]++;
    hash_table[right + 1]--;
}

int max_rating = 0;
int current = 0;
for (int i = 0; i <= 400000; i++) {
    current += hash_table[i];
    if (current > max_rating) {
        max_rating = current;
    }
}

printf("%d\n", max_rating);
return 0;

}
```

Input:

```
1
1
4
1 2 3 4
```

Output:

```
3
```